

Stat983

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1b

```
# EM-style iterative routine for the multinomial example
# Observed counts:
n1 <- 70
n2 <- 64
n3 <- 34
n4 <- 32
n  <- n1 + n2 + n3 + n4

# Function: log-likelihood as a function of theta
loglik_theta <- function(theta, n1, n2, n3, n4) {
  # ensure p_i positive
  p1 <- (1 + theta) / 3
  p2 <- (1 - theta) / 3
  p3 <- (1 + theta) / 6
  p4 <- (1 - theta) / 6
  if (any(c(p1,p2,p3,p4) <= 0)) return(-Inf)
  ll <- n1*log(p1) + n2*log(p2) + n3*log(p3) + n4*log(p4)
  return(ll)
}

# EM-style update (M-step closed form)
mstep_update <- function(n1, n2, n3, n4) {
  ( (n1 + n3) - (n2 + n4) ) / (n1 + n2 + n3 + n4)
}

# Iterative EM-like loop
em_iter <- function(n1, n2, n3, n4, theta_init = 0, tol = 1e-8, max.iter = 100) {
  theta <- theta_init
  history <- data.frame(iter = integer(0),
                        theta = numeric(0),
                        logLik = numeric(0),
                        change = numeric(0))
  for (it in 1:max.iter) {
    # E-step: here the observed aggregated cell counts are the sufficient stats,
    # so nothing to do (E-step does not change counts).
    # M-step: closed-form update
    theta_new <- mstep_update(n1, n2, n3, n4)
    ll_new <- loglik_theta(theta_new, n1, n2, n3, n4)
    change <- abs(theta_new - theta)
    history <- rbind(history, data.frame(iter = it,
```

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                                theta = theta_new,
                                logLik = ll_new, change = change))

# print progress
cat(sprintf("iter %2d: theta = %.8f, logLik = %.6f, change = %.4e\n",
            it, theta_new, ll_new, change))
if (change < tol) break
theta <- theta_new
}
return(list(theta = theta_new, history = history))
}

# run
res <- em_iter(n1, n2, n3, n4, theta_init = 0.5, tol = 1e-12, max.iter = 20)

## iter  1: theta = 0.04000000, logLik = -265.310129, change = 4.6000e-01
## iter  2: theta = 0.04000000, logLik = -265.310129, change = 0.0000e+00

res$theta

## [1] 0.04

res$history

##   iter theta    logLik change
## 1    1  0.04 -265.3101   0.46
## 2    2  0.04 -265.3101   0.00

```